

AWS IAM Users

To Begin with the Lab

Summary of the Lab

In this lab, **AWS IAM users** were introduced as the first core IAM identity used to represent a real human who needs access to AWS resources. The video showed how to create a user, give it console access, and attach **Amazon EC2 Full Access** so the user could manage EC2 resources independently of the root user. It also explained how access keys and secret access keys are used for programmatic access through the AWS CLI or SDK, and why they must be stored securely. The lab then demonstrated removing the policy to show that without an attached policy, the IAM user loses access. Finally, it covered **MFA** and password policy settings to improve user security and enforce stronger login standards.

Understanding IAM Users

An **IAM user** is a logical identity in AWS that represents a person or service account that needs access to AWS resources. It exists so the root user does not have to be shared, and permissions can be assigned more safely and specifically to each person. IAM users can sign in to the console with a password, or they can use access keys for programmatic access through the CLI or SDK. This makes IAM users useful for day-to-day AWS access, but only when they are paired with the right policies, MFA, and password rules.

Example:

A developer named User One needs to manage EC2 instances. You create an IAM user, give console access, and attach **Amazon EC2 Full Access**. The user can then start, stop, and manage EC2 instances, but only because the policy was explicitly attached to that identity.

Lab Steps

Step 1 – Create the IAM User

- Sign in to the AWS Management Console.
- Open **IAM** → **Users**.
- Click **Create user**.

Managing human user access account by account? There's a better way.

Streamline human access to AWS and cloud apps when you enable Identity Center. [Learn more](#)

[Watch how it works](#)

One-time set up for workforce user access Centrally manage access to multiple AWS accounts Provide access centrally to the cloud applications your workforce uses All with one-click access through a simple web portal

IAM users (3) [Info](#)

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

☐	User name	Path	Group	Last activity	MFA	Password age	Console last sign-in	Access key ID
☐	[redacted]	/	0	7 hours ago	-	8 days	7 hours ago	[redacted]
☐	[redacted]	/	0	6 hours ago	-	7 hours	6 hours ago	-
☐	[redacted]	/	0	1 hour ago	-	18 days	1 hour ago	-

[Delete](#) [Create user](#)

- Enter the user name: **user-1**
- Select **Provide user access to the AWS Management Console**.
- Set a console password.
- Disable password reset on next login if desired.
- Click **Next**.
- Do not attach permissions yet.

User details

User name

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , @ _ - (hyphen)

☒ **Provide user access to the AWS Management Console - optional**

Impaired or console users with Sign-in console permissions can use the same console credentials for programmatic access without the need for access keys.

Are you providing console access to a person?

User type

☐ Specify a user in Identity Center - Recommended

We recommend that you use Identity Center to provide console access to a person. With Identity Center, you can centrally manage user access to their AWS accounts and cloud applications.

☒ **I want to create an IAM user**

We recommend that you create IAM users only if you need to enable programmatic access through access keys, service-specific credentials for AWS CodeCommit or Amazon Keyspaces, or a backup credential for emergency account access.

Console password

☐ Autogenerated password

You can view the password after you create the user.

☒ **Custom password**

Enter a custom password for the user.

☐ Show password

☐ **Users must create a new password at next sign-in - Recommended**

We recommend that you create IAM users only if you need to enable programmatic access through access keys, service-specific credentials for AWS CodeCommit or Amazon Keyspaces, or a backup credential for emergency account access.

[If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user.](#)

- Click **Create user**.

Step 2 – Attach Console Permissions

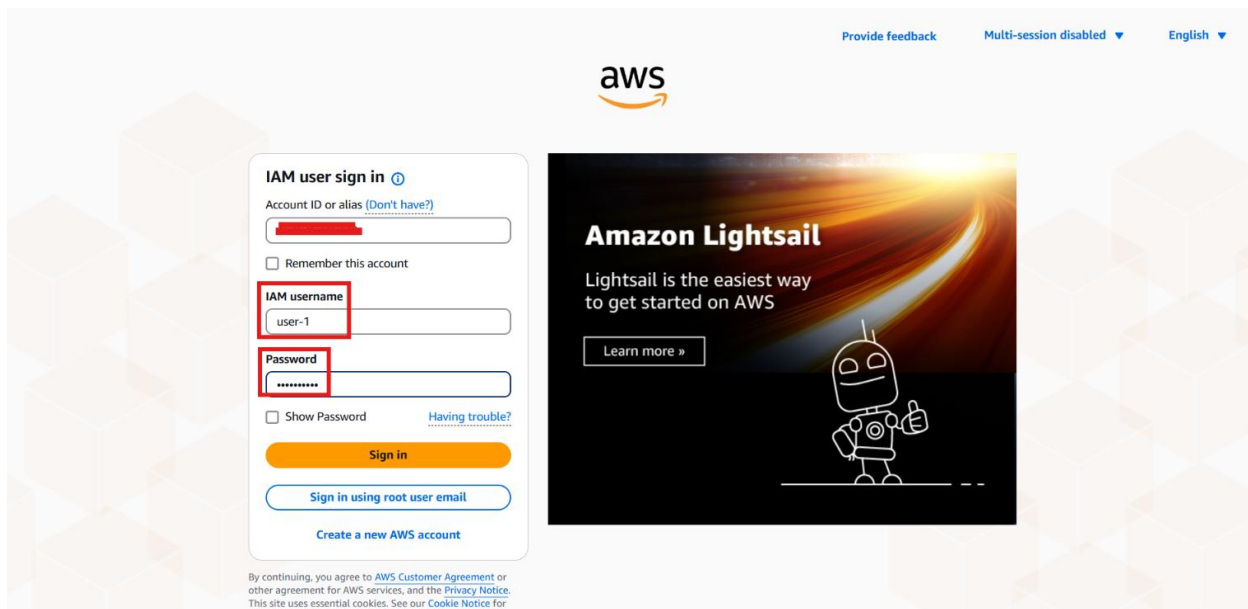
- Open the newly created IAM user.
- Click **Add permissions**.

The screenshot shows the AWS IAM console for user 'user-1'. The left sidebar contains navigation links for Identity and Access Management (IAM), Access Management, and Access reports. The main content area shows the user's summary, including their ARN, console access status, and last sign-in. Below this, there are tabs for Permissions, Groups, Tags, Security credentials, and Last Accessed. The 'Permissions' tab is active, showing a list of permissions policies. A red box highlights the 'Add permissions' button in the top right corner of the permissions section.

- Choose **Attach policies directly**.
- Select **AmazonEC2FullAccess**.
- Click **Next**.

The screenshot shows the 'Add permissions' step in the AWS IAM console. The left sidebar shows the progress: Step 1 (Add permissions), Step 2, and Review. The main content area is titled 'Add permissions' and includes a sub-header 'Permissions options'. There are three radio button options: 'Add user to group', 'Copy permissions', and 'Attach policies directly'. The 'Attach policies directly' option is selected and highlighted with a red box. Below this, there is a section for 'Permissions policies (1/2645)'. A search bar contains 'AmazonEC2F'. A table lists the policies, with 'AmazonEC2FullAccess' highlighted in blue and its name boxed in red. At the bottom right, there are 'Cancel' and 'Next' buttons, with the 'Next' button highlighted in red.

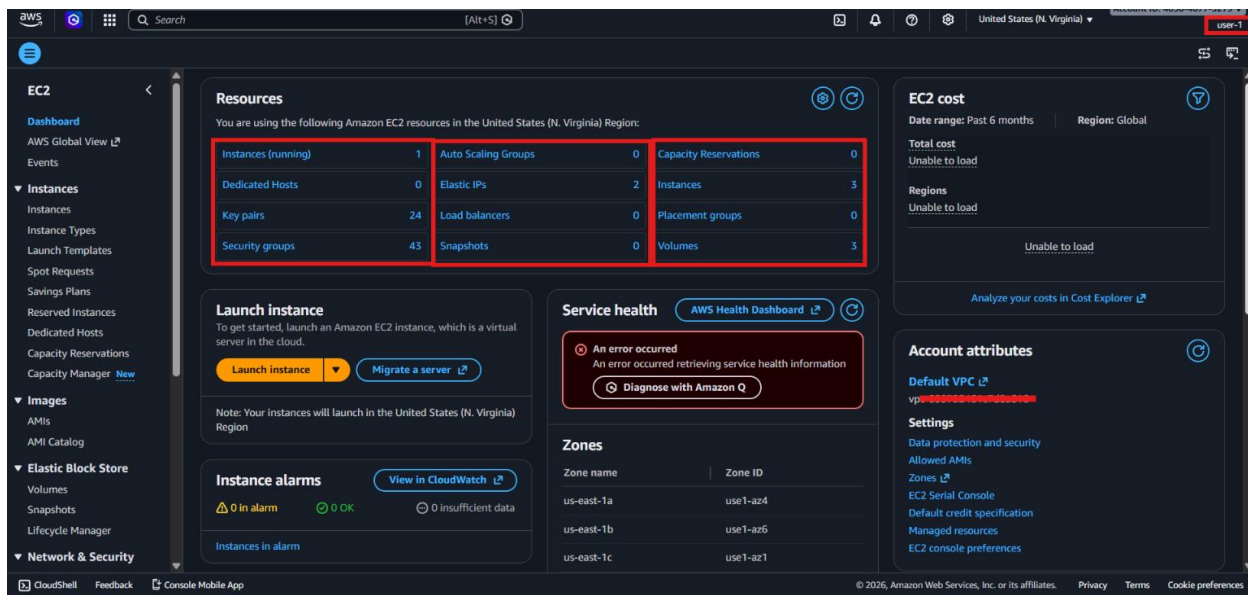
- Click **Add permissions**.
- Sign in using the IAM user URL.



- Verify that EC2 access is available.

Step 3 – Verify EC2 Access

- Open Amazon EC2.
- Confirm that the user can view EC2 resources.



- Start or stop an instance to verify the permissions.
- Observe that the user can manage EC2 because the policy was attached.

Step 4 – Create Access Keys for Programmatic Access

- Open the IAM user.
- Go to the **Security credentials** tab.
- Click **Create access key**.

The screenshot shows the AWS IAM console for user 'user-1'. The 'Security credentials' tab is active. It displays 'Access key 1' with a 'Create access key' button. Below this, the 'Access keys (0)' section also has a 'Create access key' button. The 'Console sign-in' section shows the console sign-in link and password status. The 'Multi-factor authentication (MFA) (0)' section shows no MFA devices are assigned.

- Choose **AWS CLI** as the use case.
- Confirm the warning.

The screenshot shows the 'Create access key' wizard in the AWS IAM console. The 'Command Line Interface (CLI)' use case is selected. The 'Alternatives recommended' section lists recommended alternatives. The 'Confirmation' section has a checkbox that is checked.

- Add a tag if needed.
- Click **Create access key**.

Step 1: Access key best practices & alternatives

Step 2 - optional: Set description tag

Step 3: Retrieve access keys

Set description tag - optional

The description for this access key will be attached to this user as a tag and shown alongside the access key.

Description tag value
 Describe the purpose of this access key and where it will be used. A good description will help you rotate this access key confidently later.

Maximum 256 characters. Allowed characters are letters, numbers, spaces representable in UTF-8, and: _ . : / * + = @

Cancel Previous **Create access key**

- Download or save the access key and secret access key immediately.
- Store the secret key securely because it will not be shown again.

© This is the only time that the secret access key can be viewed or downloaded. You cannot recover it later. However, you can create a new access key any time.

Step 1: Access key best practices & alternatives

Step 2 - optional: Set description tag

Step 3: Retrieve access keys

Retrieve access keys

Access key
 If you lose or forget your secret access key, you cannot retrieve it. Instead, create a new access key and make the old key inactive.

Access key

Secret access key
 Show

Access key best practices

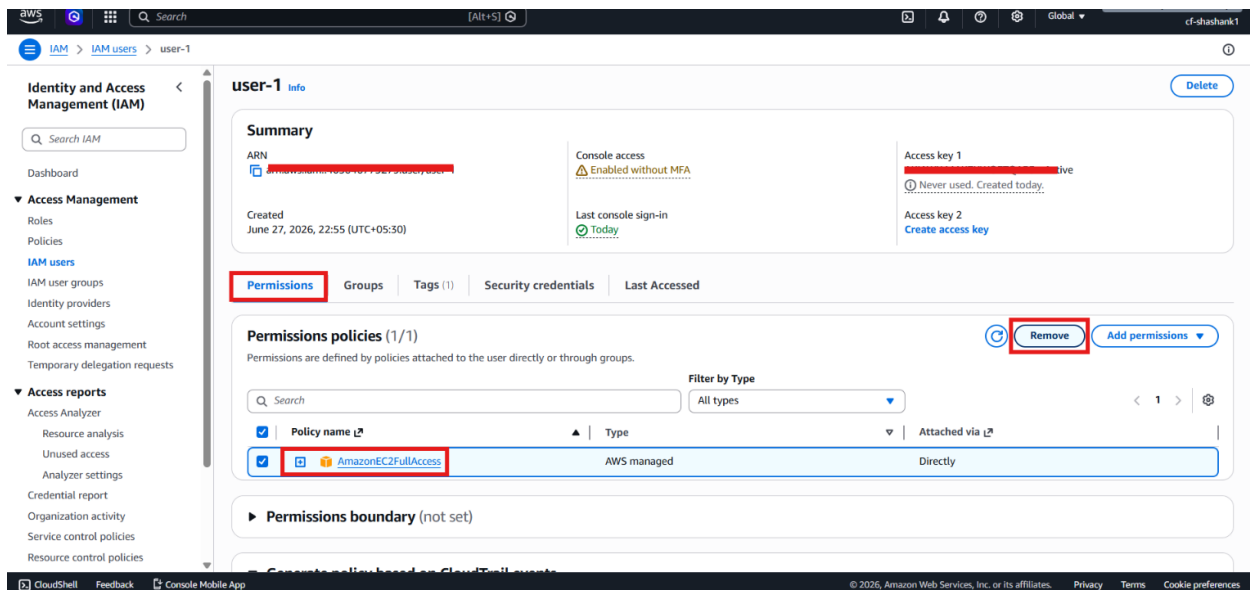
- Never store your access key in plain text, in a code repository, or in code.
- Disable or delete access key when no longer needed.
- Enable least-privilege permissions.
- Rotate access keys regularly.

For more details about managing access keys, see the [best practices for managing AWS access keys](#).

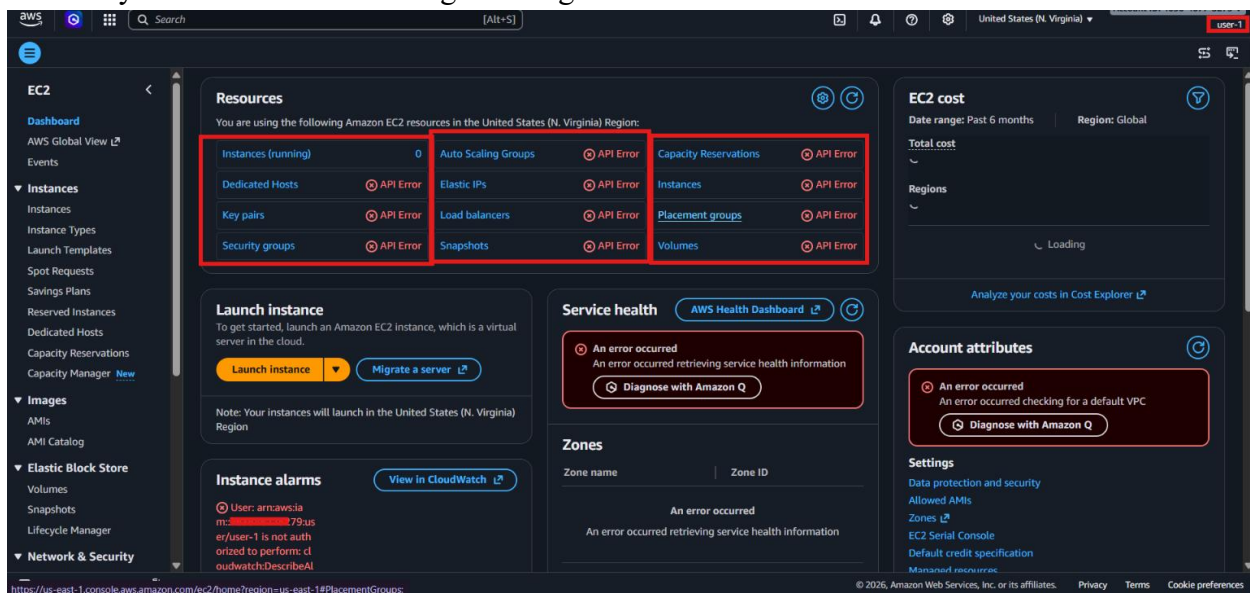
Download .csv file Done

Step 5 – Remove the Policy and Observe Access Loss

- Return to the IAM user permissions.
- Detach **AmazonEC2FullAccess**.



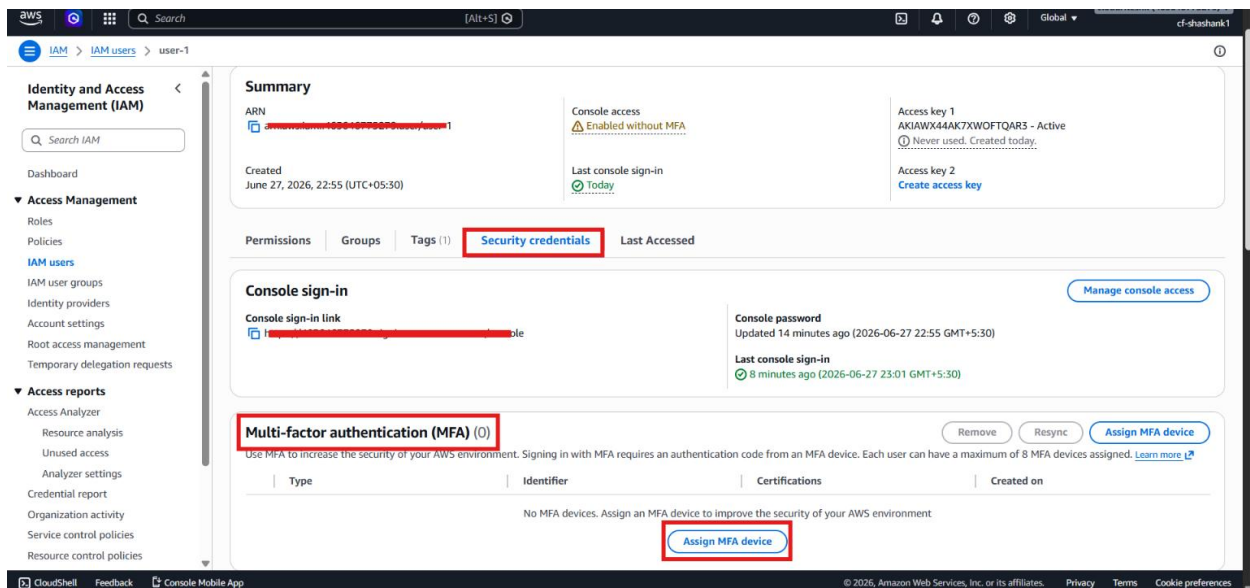
- Refresh the console.
- Verify that the user can no longer manage EC2 resources.



- Confirm that IAM users must have explicit policy permissions to access AWS services.

Step 6 – Enable MFA

- Open the IAM user.
- Go to **Security credentials**.
- Scroll to **Multi-factor authentication**.
- Click **Assign MFA device**.



- Choose an MFA method such as an authenticator app.
- Scan the QR code with Google Authenticator or a similar app.
- Enter the verification codes.
- Complete MFA setup.

Verification

- The IAM user was created successfully.
- Console access was configured.
- **Amazon EC2 Full Access** allowed the user to manage EC2 resources.
- Access keys were created for programmatic access.
- Removing the policy removed EC2 access.
- MFA setup was reviewed.